

Year 2: Development of R.A.M.P.A.T. (Rapid, Affordable, Modular, Printed, Aerial Tiltrotor) For Emergency Response

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INTRODUCTION

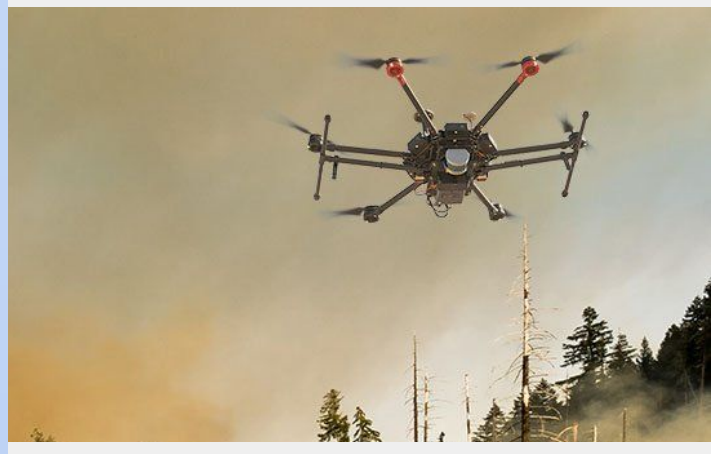
Background



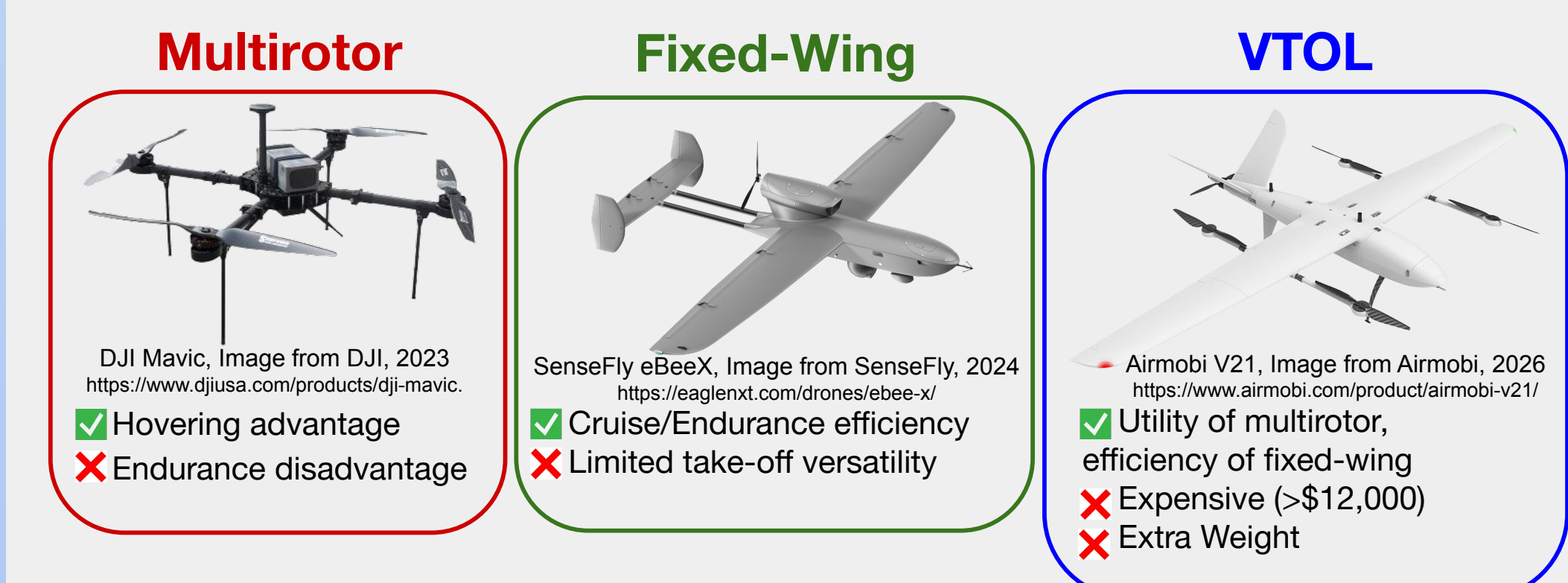
Aftermath of Hurricane Melissa in Jamaica, 2025. Floods destroyed the country's ground infrastructure, leading to survivor searching efforts taking nearly two months. Source: <https://supportjamaica.gov.jm/>

Unmanned Aerial Vehicles (UAV)

An Unmanned Aerial Vehicle (UAV) is any vehicle that doesn't have a pilot on board and is either remotely piloted or autonomous.



UAV in a California forest fire, 2024. Source: www.fire.ca.gov



RESEARCH OBJECTIVES

Engineering Problem

Current UAVs are not sufficiently accessible to global first responder groups due to a **lack of balance between performance and monetary cost**.

Research Question

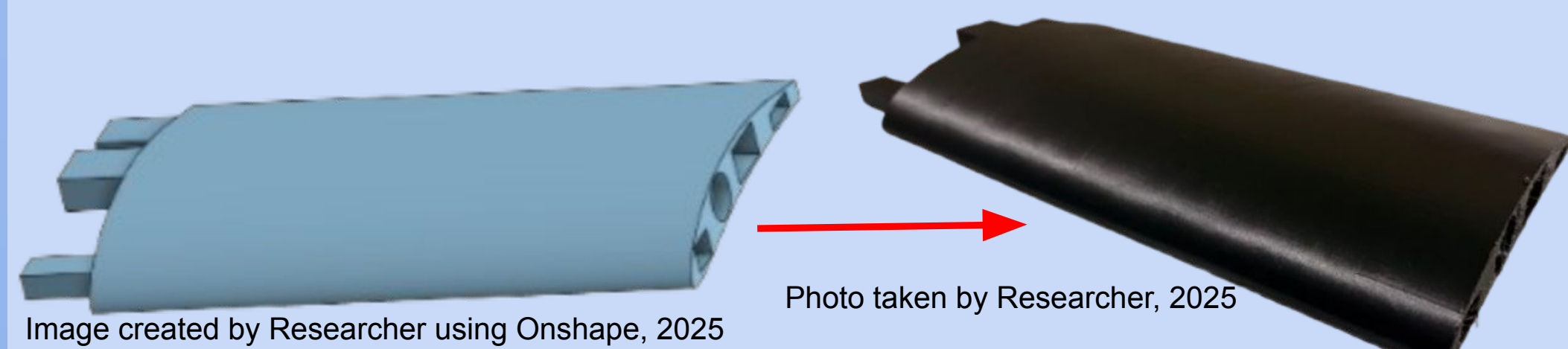
Can we develop a cost-efficient UAV that balances performance and cost to be an effective tool for first responders?

Engineering Design Criteria

Performance: >1800 acre search capability, >2.75 hour endurance	Affordability: Contain a materials cost of <\$1000	Functionality: Swappable parts, VTOL capability
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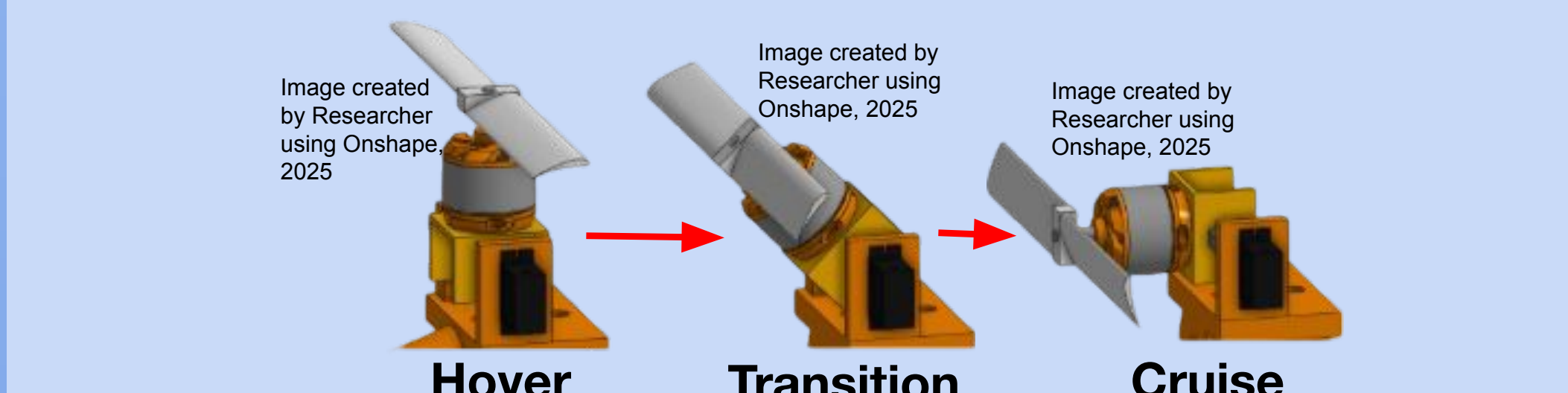
Brainstorming R.A.M.P.A.T.

3D Printed Manufacturing



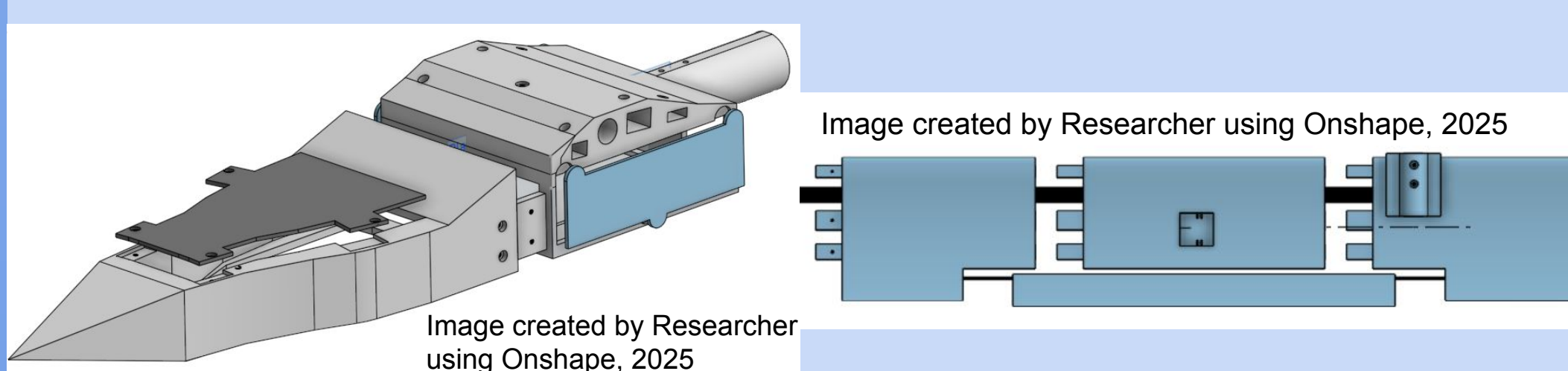
- CAD Model**
- 3D Print**
- PETG (Polyethylene Terephthalate Glycol-modified) enables rapid, low-cost airframe production
- Allows **lightweight, strong structures** to maximize flight endurance for long-range search.
- UV-resistant & waterproof**

Tiltrotor Propulsion



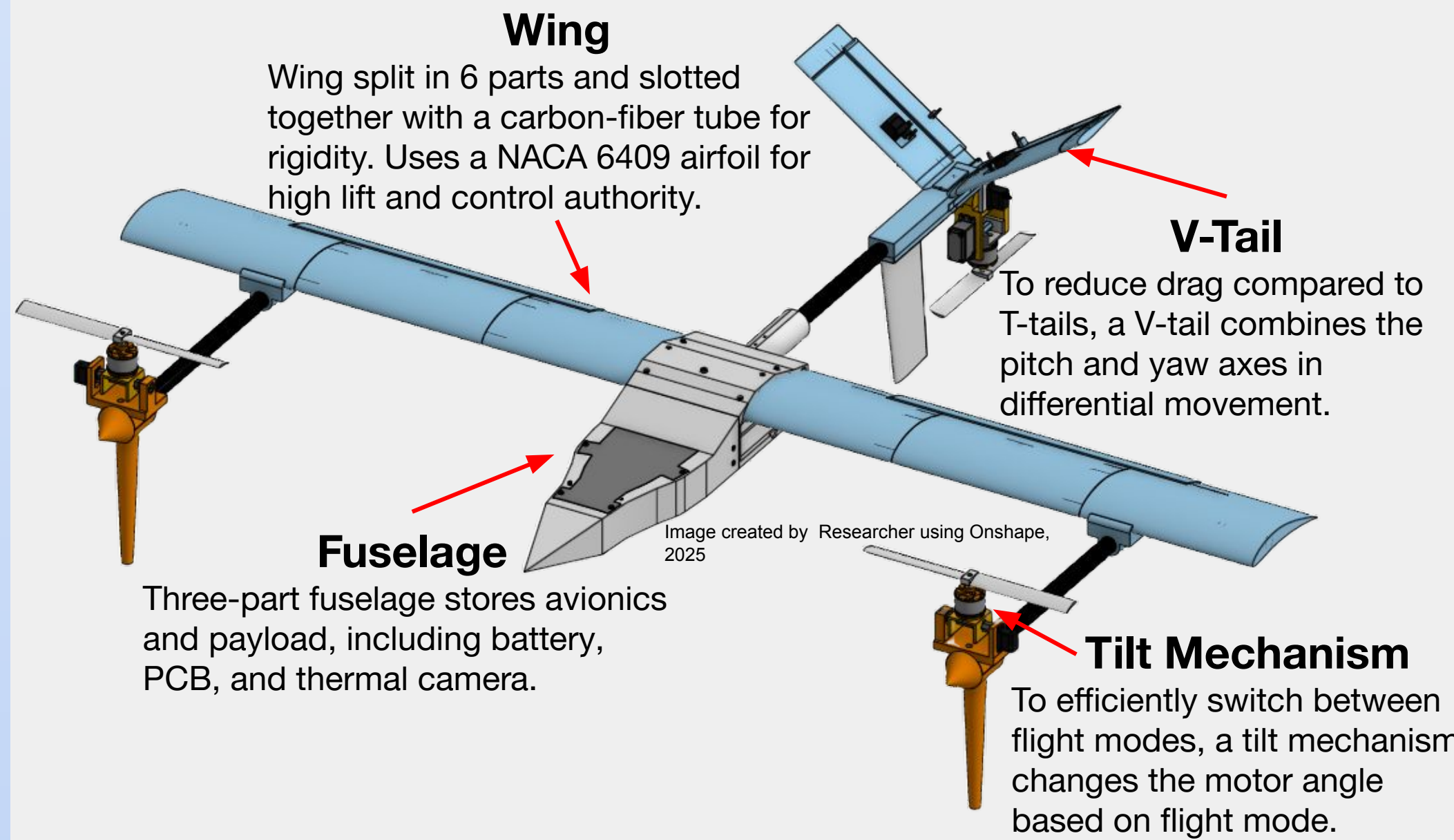
- Traditional VTOLs use **redundant propulsion systems** for hover and cruise.
- Tiltrotor VTOLs **combine hover and cruise** in the same propulsion system, reducing cost.
- While developed, tiltrotor VTOLs are rarely used in first response and have never been implemented in a 3D-printed first response aircraft.

Modular Design

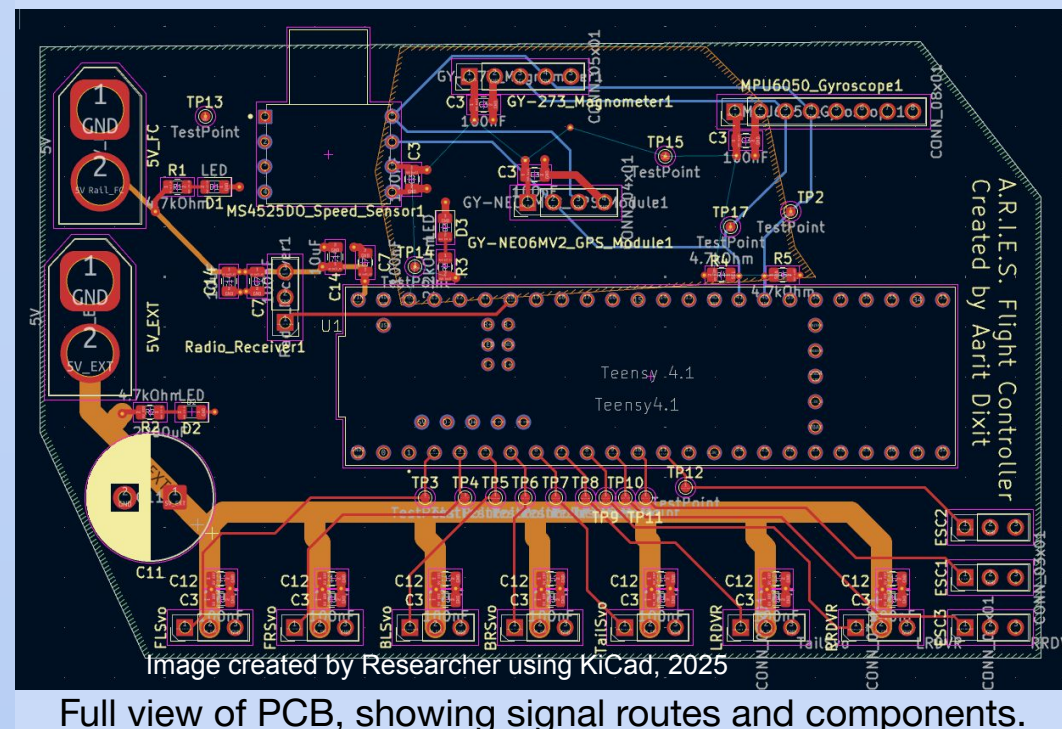


- Parts slot together, allowing for **repairs and upgrades**.
- Combined with 3D printing, this allows R.A.M.P.A.T. to be **easily repaired**.
- Total assembly time (including manufacturing+part sourcing) **<48 hours, 5x less than industry UAVs**.

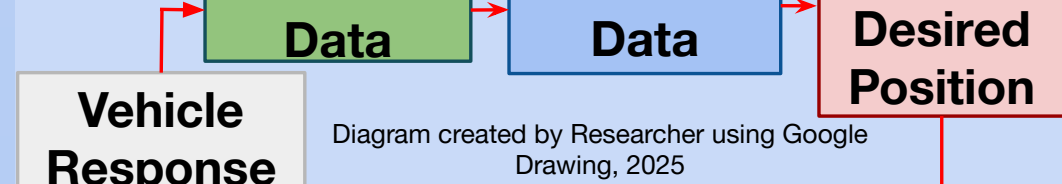
METHODS



Flight Controller



Full view of PCB, showing signal routes and components.



Flight control loop, controlling stabilization and movement.

Proportion-Integral-Derivative Equation

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

Equation created by Researcher using Overleaf, 2025

- where:
- $u(t)$ is the actuator output command.
 - $e(t)$ is the error.
 - K_p , K_i , and K_d are the proportion, integral, and derivative coefficients.

Due to unique design and cost constraints, a **custom flight controller PCB** was developed (diagram to the left).

Key features:

- > **85% cost reduction (\$75 vs. \$500+)**
- > **Full sensor integration**
- > **Long-range radio telemetry**
- > **High-frequency flight stabilization**
- > **Off-the-shelf sensors with custom control software**

Transition Control Mixing Equation

$$\delta = \min(1, \max(0, \frac{V - 10}{5})) \cdot \delta_{cmd}$$

Equation created by Researcher using Overleaf, 2025

- where:
- δ is the true control surface output.
 - δ_{cmd} is the pilot-given control surface output.
 - V represents the measured velocity

Computational Fluid Dynamics

Geometry 1

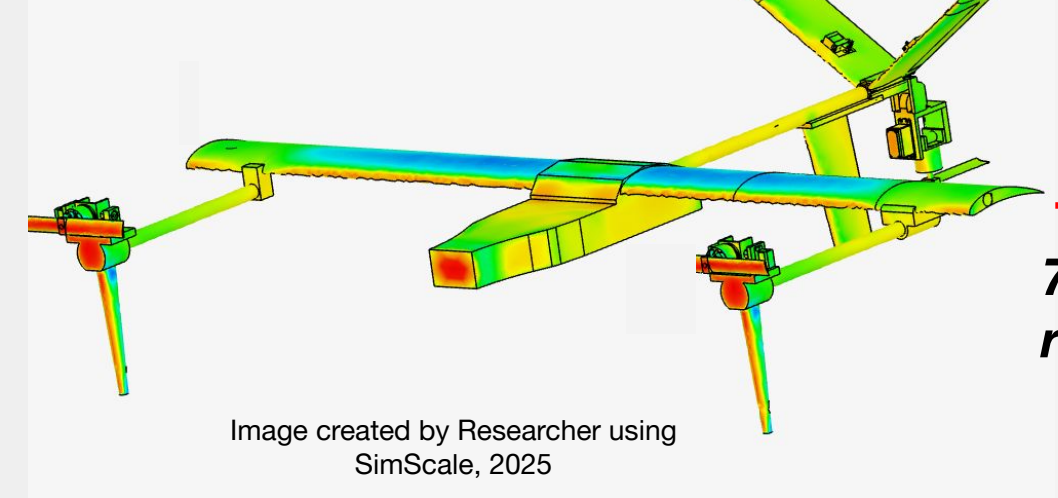


Image created by Researcher using SimScale, 2025

- Geometric Optimization:**
- > Reducing high-pressure (red) areas.
 - > Increasing low-pressure (blue) areas on top of the wing.
 - > These two factors increase Lift/Drag ratio.

Mesh Independence Study

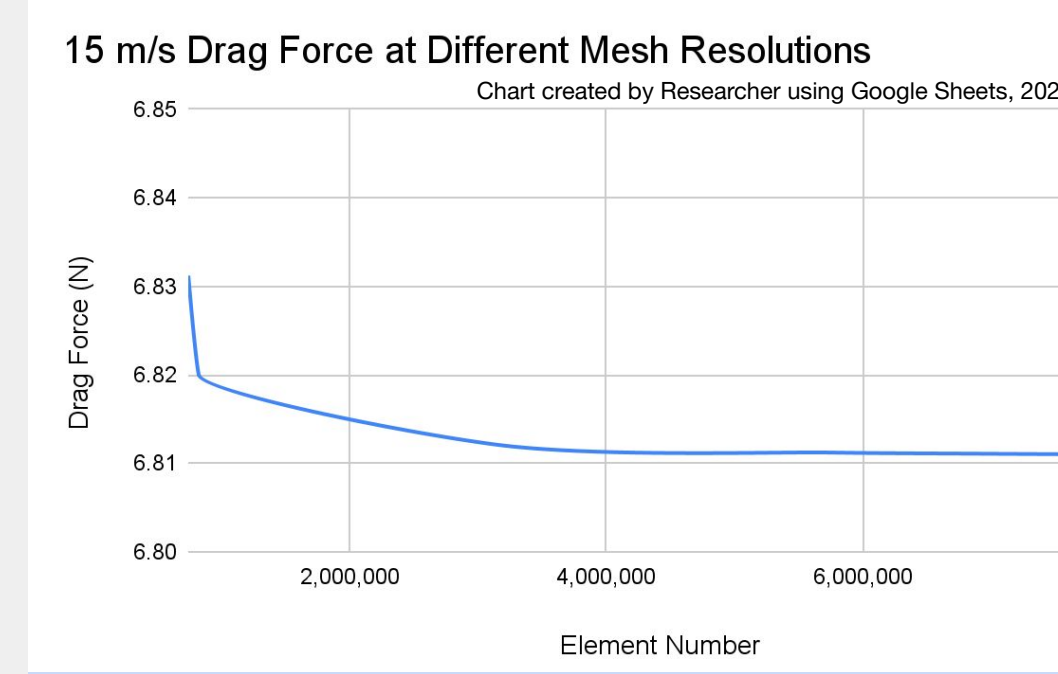


Chart created by Researcher using Google Sheets, 2026

Geometry 7

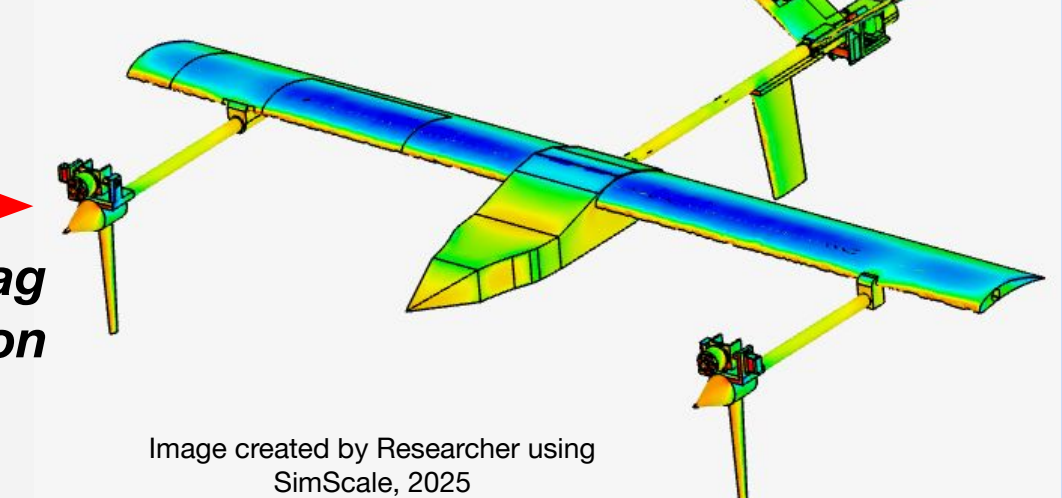


Image created by Researcher using SimScale, 2025

- Final Results:**
- > L/D ratio of 12.1 (industry is 10+).
 - > High lift at lower speeds, increasing payload capacity.
 - > Streamlined fuselage + Landing gear.

Residual Convergence

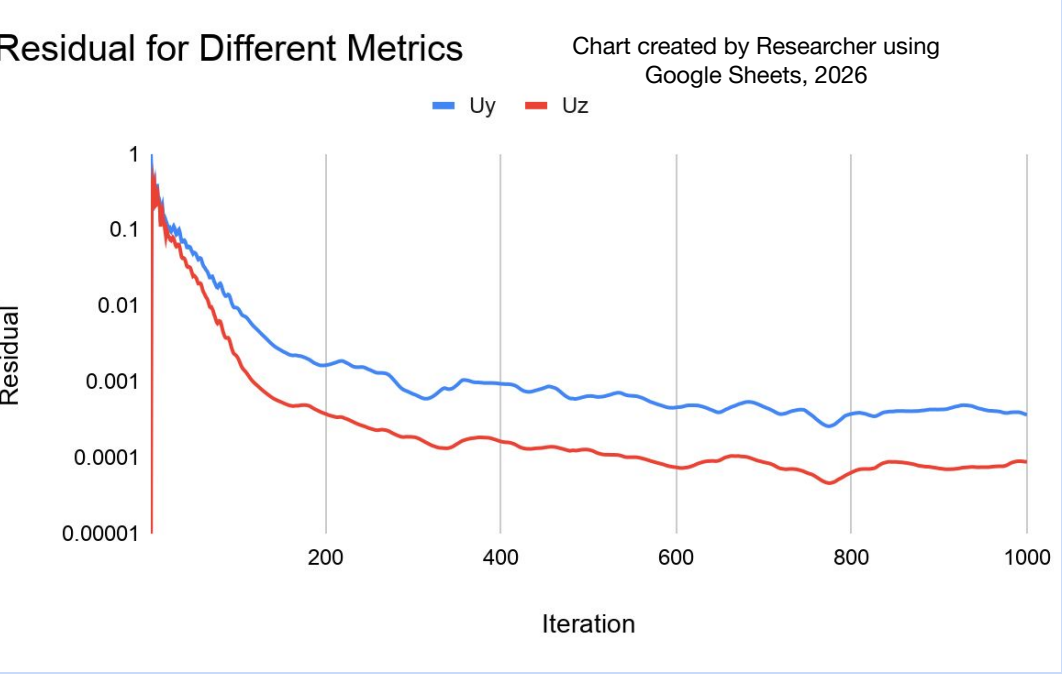
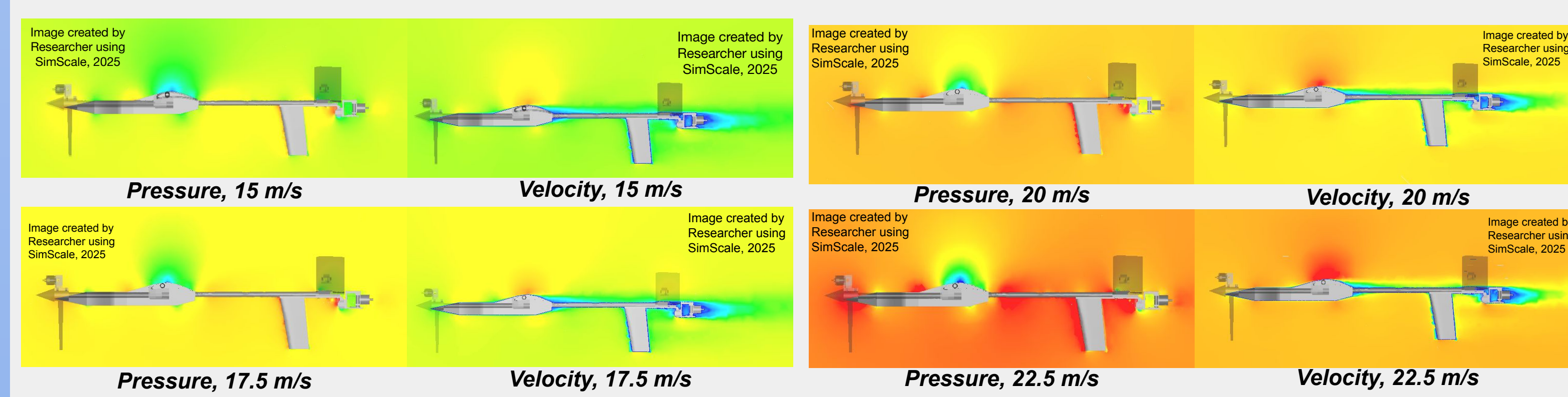


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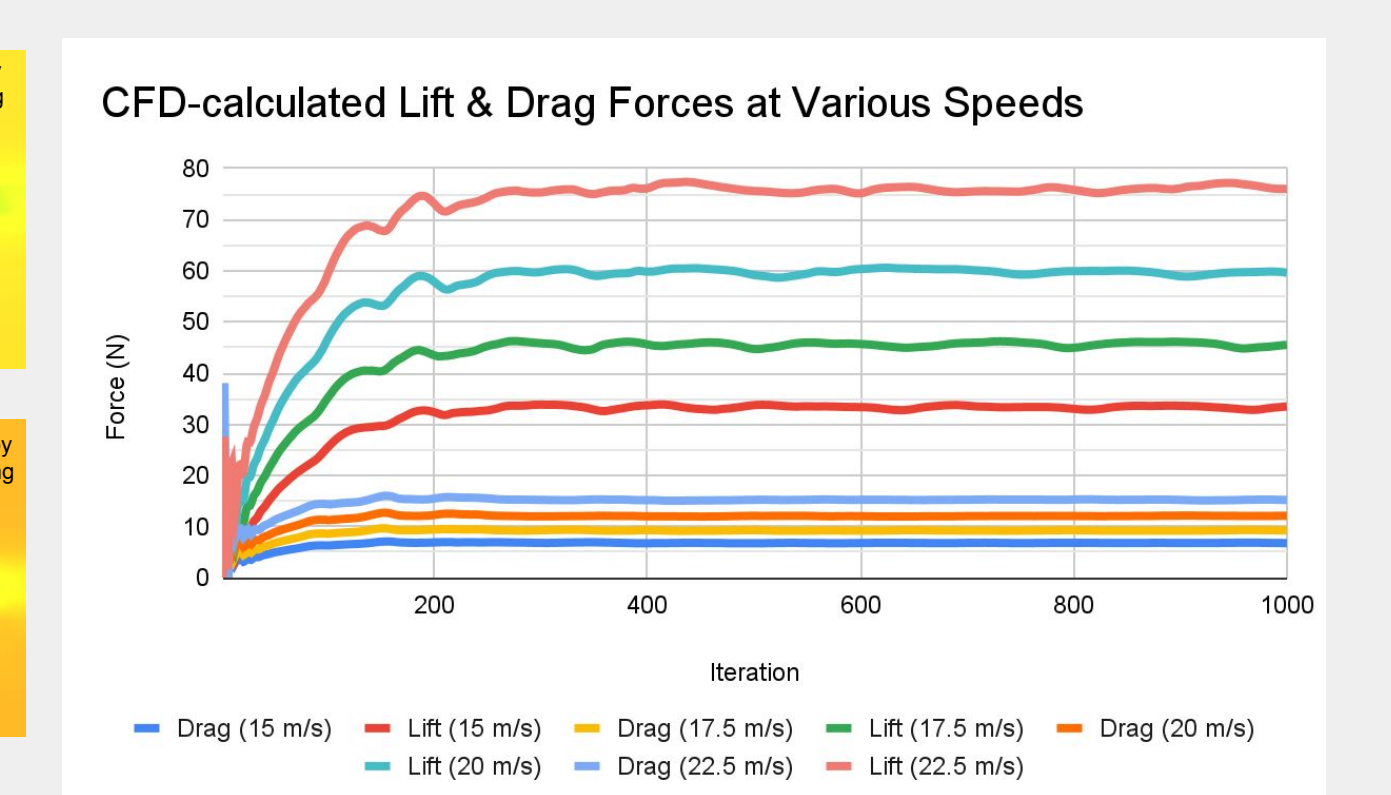
RESULTS

Fig 1. Airflow Property Fields of R.A.M.P.A.T. Geometry



Images of CFD fields, highlighting optimal aerodynamics and airflow separation.

Fig 2. CFD-Calculated Forces



CFD forces show convergence, highlighting a valid simulation.

Chart created by Researcher using Google Sheets, 2026

Fig 3. Data Extraction Process

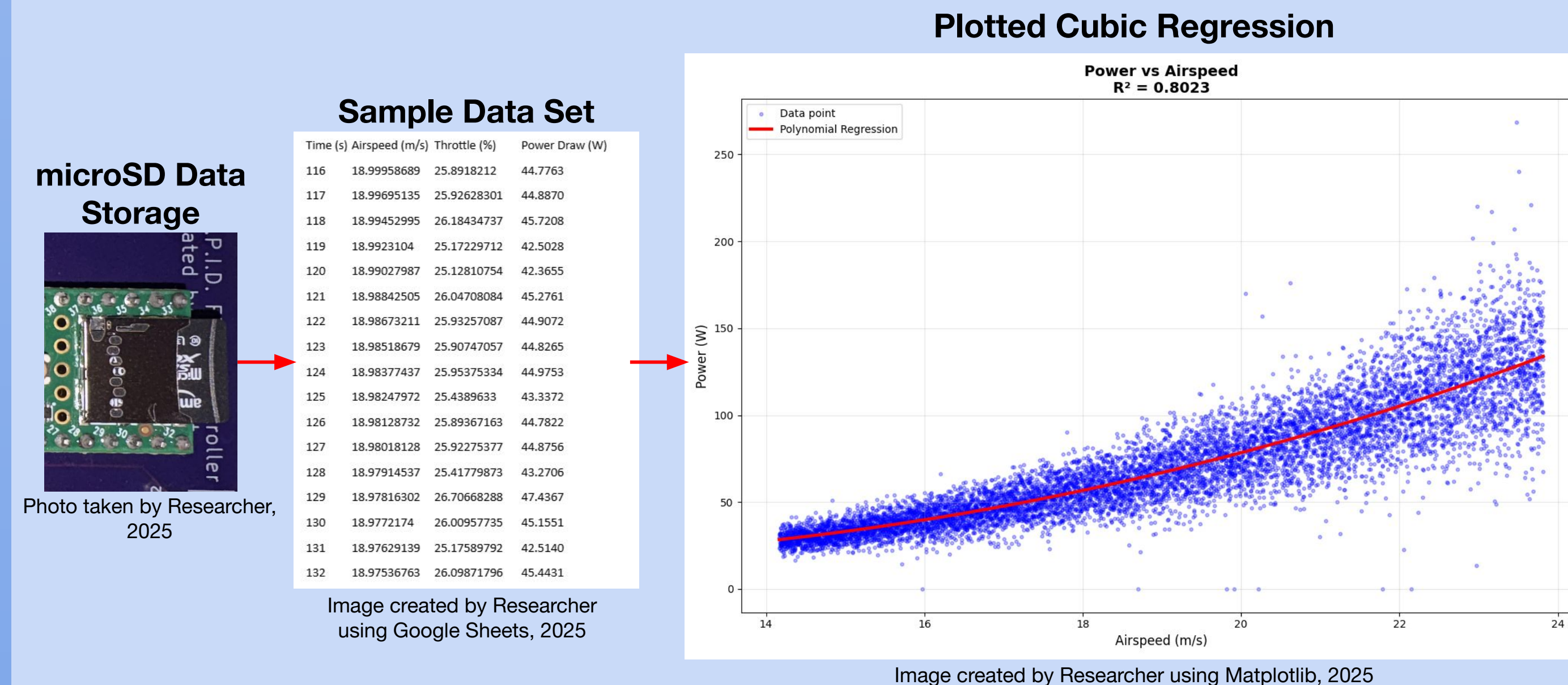
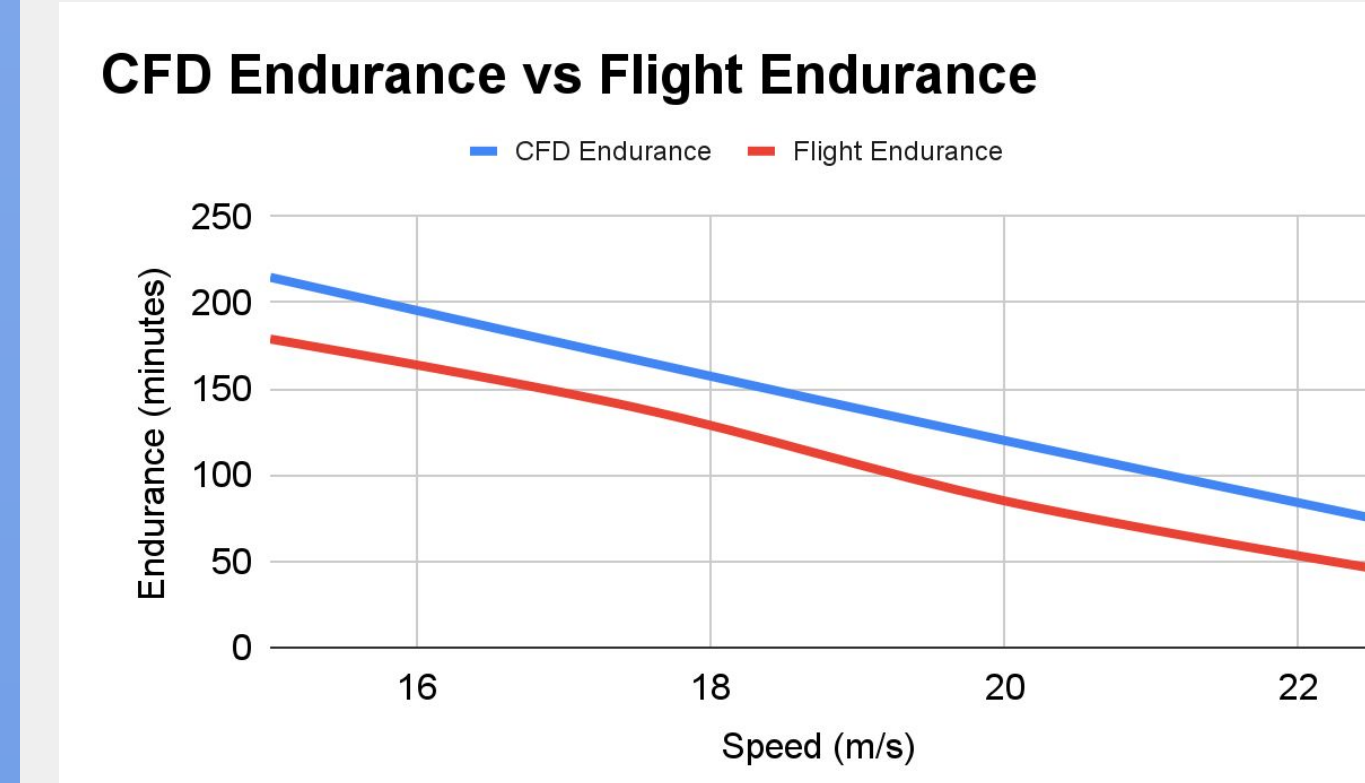


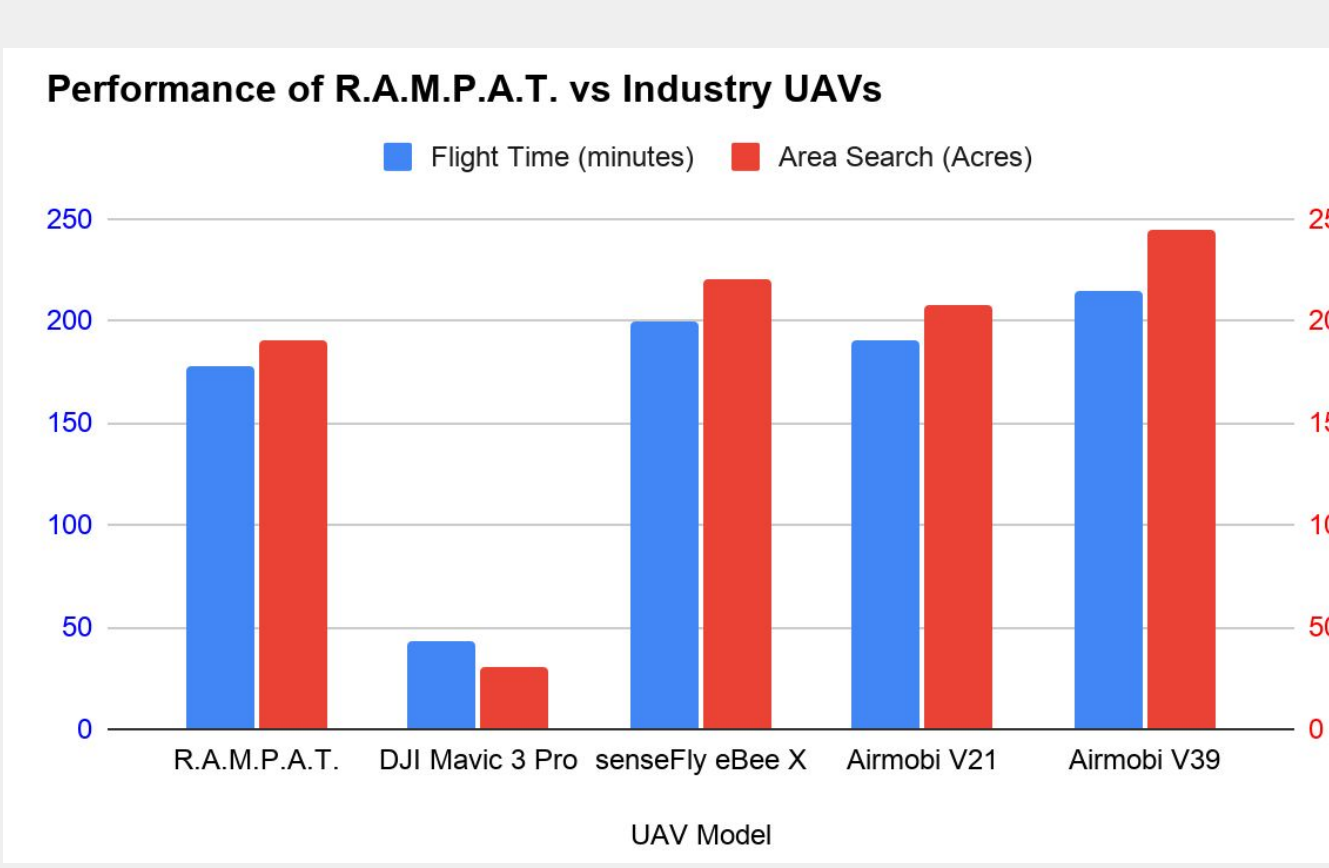
Fig 5. R.A.M.P.A.T. Flight Endurance Compared with CFD-calculated Flight Time



Flight tests come close to the CFD-calculated "maximum" flight time.

Chart created by Researcher using Google Sheets, 2026

Fig 6. R.A.M.P.A.T. Performance vs Industry



R.A.M.P.A.T. surpasses its cost category while being comparable to \$10,000 systems in the key metrics of endurance and search capability.

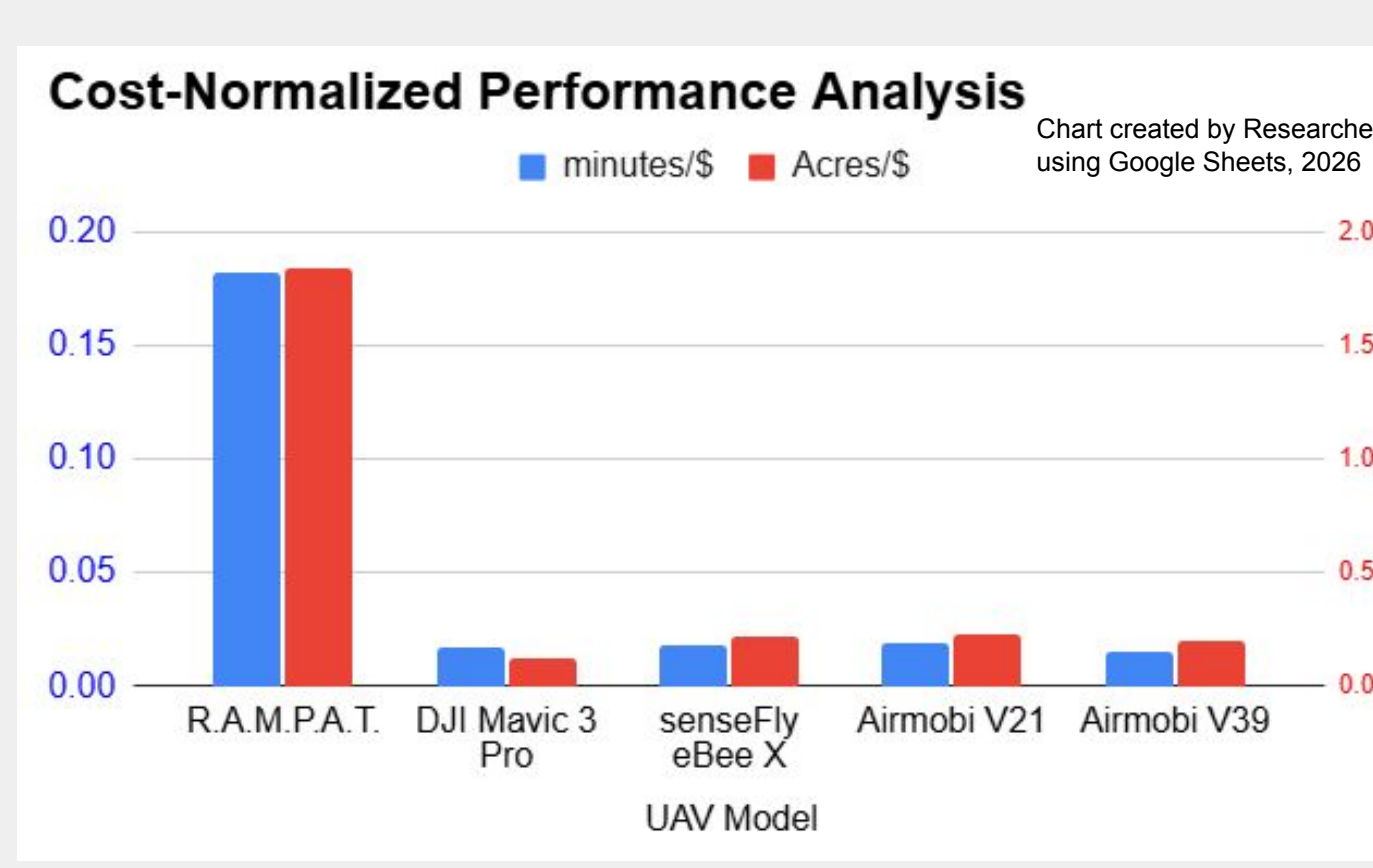
Chart created by Researcher using Google Sheets, 2026

Fig 4. Industry Comparison Table

UAV	Cost-Flight Time (Minutes)	Optimal Area Search (Acres)	Cost (USD)
R.A.M.P.A.T.	178	1900	\$978
DJI Mavic 3 Pro	43	300	\$2,500
senseFly eBeeX	200	2200	\$10,000
Airmobi V21	190	2075	\$10,000
Airmobi V39	215	2450	\$11,000

Table created by Researcher using Google Sheets, 2026

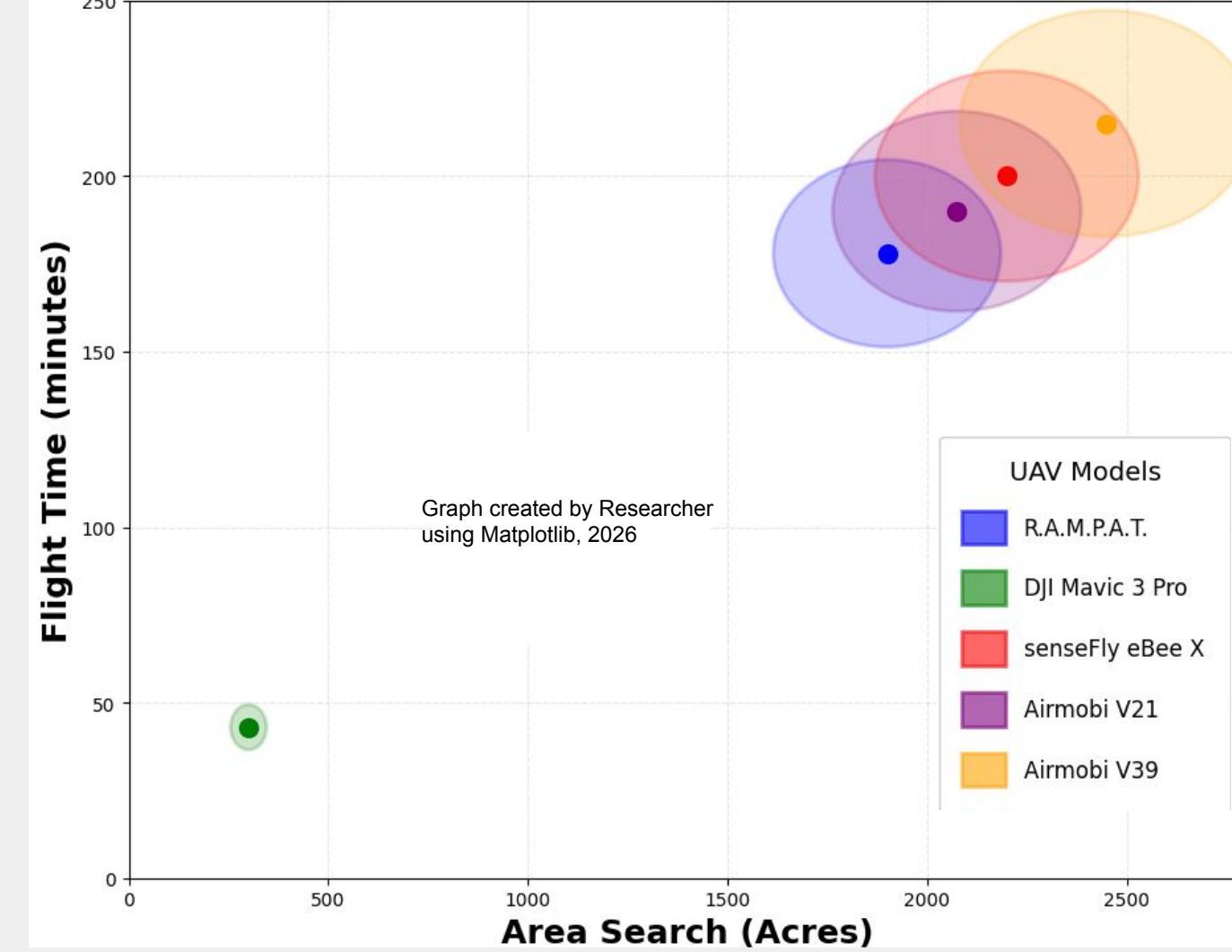
Fig 7. Cost-Normalized Performance of R.A.M.P.A.T. vs Existing UAVs



R.A.M.P.A.T. has an order-of-magnitude advantage in the metrics of Flight Time Per Dollar and Search Area Per Dollar (Based on retail price of UAVs/material cost of R.A.M.P.A.T., as electrical cost is negligible when environmental factors are taken to consideration)

STATISTICAL ANALYSIS

UAV Performance with Error Propagation



Graph displays the overlap in area search capability and flight time shown by R.A.M.P.A.T. and industry-level drones, showcasing similar performance at a fraction of the cost.

Modeling: Using manufacturer data, assorted industry UAVs were compared to R.A.M.P.A.T., with the company-reported data ranges being used for analysis.

Overlap Interpretation: Overlap of uncertainty envelopes indicates plausible performance similarity, while non-overlap suggests likely separation.

Comparative Inference: Within uncertainty bounds, R.A.M.P.A.T. overlaps with mid-tier platforms (~\$8-12k) at a cost of less than \$1000 and demonstrates clear separation from lower-cost alternatives (4-6x better), showing dominance in cost-constrained regimes.

DISCUSSION

Aerodynamic Modeling & Validation

- > CFD solutions converged across all flight regimes, confirming stable, physically valid aerodynamics.
- > Lift, drag, and power predictions aligned closely with flight-test data.
- > Cruise endurance measurements fell within CFD-predicted bounds, validating the modeling approach.

Performance Comparison

- > Commercial UAV benchmarks used manufacturer-reported ranges due to lack of variance data.
- > R.A.M.P.A.T. variance derived from repeated testing, producing tighter uncertainty (12%).
- > Performance judged by overlap of uncertainty envelopes, not single values.
- > R.A.M.P.A.T. overlaps mid-tier (\$8k-\$12k) UAVs in endurance and search capability; clearly outperforms low-cost multirotors.

Cost-Normalized Performance

- > Delivers order-of-magnitude gains in flight time per dollar and search area per dollar.
- > Advantages stem from modular 3D-printed structure, shared hover/cruise propulsion, and low-cost custom controller.

System Design and Operational Impact

- > Additive manufacturing introduces minor structural tradeoffs but maintains low failure risk (<1%).
- > Modular components enable rapid field repairs.
- > Establishes a new cost-performance benchmark for emergency-response UAVs.
- > Expands access for the 80%+ of first-responder agencies lacking aerial support.

CONCLUSIONS

- > **First 3D-printed tiltrotor VTOL** for disaster relief delivers **\$10k-class endurance** and search performance for under \$1k, including thermal imaging for full day/night missions.

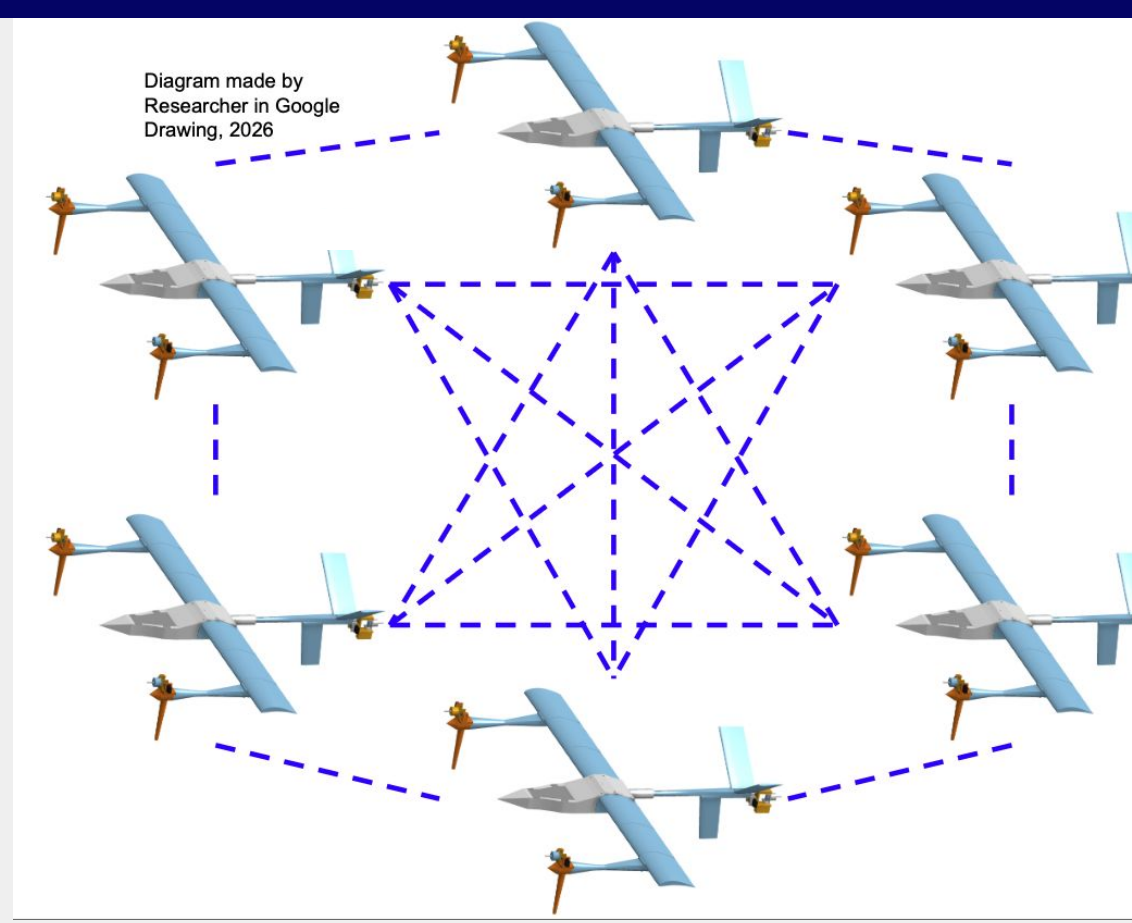
- > **Fully modular, 3D-printed airframe** enables **fast, low-cost field repairs and upgrades**, turning crashes into quick recoveries and boosting mission flexibility.

- > **Engineered for resilience with <1% failure risk**, trading minor impact-resistance limits for **unmatched maintainability and continuous deployment** for fire departments, NGOs, and community teams.

Overall, the engineering design criteria were met based on flight data, validating R.A.M.P.A.T.'s potential as a **viable solution for global first responders**.

FUTURE WORK

- > **Real-World Mission Capability**
- > **Advancing System Autonomy**
- > **Increasing Deployment Scalability**
- > **Further Environmental and Operational Testing**



Example swarm network, with 6 R.A.M.P.A.T. UAVs connected with each other to scan an area efficiently.